1. Name of the Student

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2. Title

Timer-Based LED Flasher PCB (Assignment Question 6)

Design overview

A signage company needs a simple circuit that periodically flashes an LED to attract attention on information displays and warning signs. The design uses an NE555 timer in astable mode driving an LED through a current-limiting resistor, powered from a 9 V supply via a 2-pin connector. The board was designed in KiCad (schematic, footprint assignment, layout and routing).

Frequency calculation

$f = 1.44 / ((R1 + 2{\cdot}R2) \cdot C1) = 1.44 / ((1\ k\Omega + 2 \times 47\ k\Omega) \times 10\ \mu F) = 1.44 / 0.95 \approx 1.5$ Hz

High time $t_H = 0.693\ (R1 + R2)\ C1 \approx 0.33$ s; low time $t_L = 0.693\ R2\ C1 \approx 0.33$ s; duty cycle ≈ 51%.

LED current: $I_{OUT} = (V_{OUT} - V_F) / R3 \approx (9 - 1.5 - 2.0) / 470 \approx 12$ mA (safe for a standard 5 mm LED).

3. Schematic

[KiCad Schematic Visualization Placeholder]

Netlist / connections

Net	Connections
+9V	J1-1, U1 pin 8 (VCC), U1 pin 4 (RESET), R1
GND	J1-2, U1 pin 1, C1(-), C2, D1 cathode
N_DISCH	R1, R2, U1 pin 7 (DISCH)
N_TIMING	R2, U1 pin 6 (THRES), U1 pin 2 (TRIG), C1(+)
N_CTRL	U1 pin 5 (CTRL), C2
N_OUT	U1 pin 3 (OUT), R3
N_LED	R3, D1 anode

4. Screenshot of Footprint After Routing Is Completed

[KiCad Routed Footprint View Placeholder]

Footprint assignment

Ref	Footprint
U1	Package_DIP:DIP-8_W7.62mm
R1, R2, R3	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
C1	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
C2	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
D1	LED_THT:LED_D5.0mm
J1	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical

5. 2D PCB View

6. 3D PCB View

7. Bill of Materials (BOM)

Ref	Qty	Value	Footprint	Description
U1	1	NE555	DIP-8_W7.62mm	555 timer IC
R1	1	1 k	R_Axial_DIN0207 P10.16	Timing resistor
R2	1	47 k	R_Axial_DIN0207 P10.16	Timing resistor
R3	1	470	R_Axial_DIN0207 P10.16	LED current limiting
C1	1	10 uF	CP_Radial_D5.0mm_P2.00mm	Timing capacitor (electrolytic)
C2	1	10 nF	C_Disc_D5.0mm_P5.00mm	Control-pin bypass
D1	1	LED red	LED_D5.0mm	Flasher LED
J1	1	Conn_01x02	PinHeader_1x02_P2.54mm	9 V power input

Total components: 9 (8 line items).